AI Driven Full Stack Development (AI308B)

Moodle MSE2 Submission Format

Name

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Branch

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Section

C

Shift

2nd

Case Study Name

Student Grievance Management System

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1. GitHub Repository Link

https://github.com/UpwanSingh/AIFSD_MSE_2

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2. Render Deployment Links for all Routes

Backend Deployment Link

https://student-grievance-backend-mfqb.onrender.com

Frontend Deployment Link

https://student-grievance-frontend-od85.onrender.com

Backend API Routes (Live)

- https://student-grievance-backend-mfqb.onrender.com/api/register
- https://student-grievance-backend-mfqb.onrender.com/api/login
- https://student-grievance-backend-mfqb.onrender.com/api/grievances
- https://student-grievance-backend-mfqb.onrender.com/api/grievances/:id
- https://student-grievance-backend-mfqb.onrender.com/api/grievances/search?title=xyz

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3. Project Code

Backend Code

Index.js (server.js)

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const dotenv = require("dotenv");

const authRoutes = require("./routes/authRoutes");
const grievanceRoutes = require("./routes/grievanceRoutes");

dotenv.config();

const app = express();

app.use(cors());
app.use(express.json());

app.get("/", (req, res) => {
  res.json({ message: "Student Grievance Management API is running" });
});

app.use("/api", authRoutes);
app.use("/api", grievanceRoutes);

const PORT = process.env.PORT || 5000;

const startServer = async () => {
  try {
    await mongoose.connect(process.env.MONGO_URI);
    console.log("MongoDB connected");

    app.listen(PORT, () => {
      console.log(`Server running on port ${PORT}`);
    });
  } catch (error) {
    console.error("Failed to connect MongoDB:", error.message);
    process.exit(1);
  }
};

startServer();
```

Student Model (model.js equivalent)

```
const mongoose = require("mongoose");

const studentSchema = new mongoose.Schema(
  {
    name: {
      type: String,
      required: true,
```

```
      trim: true,
    },
    email: {
      type: String,
      required: true,
      unique: true,
      lowercase: true,
      trim: true,
    },
    password: {
      type: String,
      required: true,
      minlength: 6,
    },
  },
  { timestamps: true }
);

module.exports = mongoose.model("Student", studentSchema);
```

Grievance Model

```
const mongoose = require("mongoose");

const grievanceSchema = new mongoose.Schema(
  {
    title: {
      type: String,
      required: true,
      trim: true,
    },
    description: {
      type: String,
      required: true,
      trim: true,
    },
    category: {
      type: String,
      enum: ["Academic", "Hostel", "Transport", "Other"],
      required: true,
    },
    date: {
      type: Date,
      default: Date.now,
    },
    status: {
      type: String,
      enum: ["Pending", "Resolved"],
      default: "Pending",
    },
    student: {
      type: mongoose.Schema.Types.ObjectId,
      ref: "Student",
      required: true,
    },
  },
```

```
  { timestamps: true }
);

module.exports = mongoose.model("Grievance", grievanceSchema);
```

Auth Routes

```
const express = require("express");
const bcrypt = require("bcryptjs");
const jwt = require("jsonwebtoken");
const Student = require("../models/Student");

const router = express.Router();

router.post("/register", async (req, res) => {
  try {
    const { name, email, password } = req.body;

    if (!name || !email || !password) {
      return res.status(400).json({ message: "All fields are required" });
    }

    const existingStudent = await Student.findOne({ email });
    if (existingStudent) {
      return res.status(409).json({ message: "Duplicate email" });
    }

    const hashedPassword = await bcrypt.hash(password, 10);

    const student = await Student.create({
      name,
      email,
      password: hashedPassword,
    });

    return res.status(201).json({
      message: "Registration successful",
      student: {
        id: student._id,
        name: student.name,
        email: student.email,
      },
    });
  } catch (error) {
    return res.status(500).json({ message: "Server error", error: error.message });
  }
});

router.post("/login", async (req, res) => {
  try {
    const { email, password } = req.body;

    if (!email || !password) {
      return res.status(400).json({ message: "Email and password are required" });
    }
```

```
    const student = await Student.findOne({ email });
    if (!student) {
      return res.status(401).json({ message: "Invalid login" });
    }

    const isMatch = await bcrypt.compare(password, student.password);
    if (!isMatch) {
      return res.status(401).json({ message: "Invalid login" });
    }

    const token = jwt.sign(
      {
        id: student._id,
        name: student.name,
        email: student.email,
      },
      process.env.JWT_SECRET,
      { expiresIn: "1d" }
    );

    return res.status(200).json({
      message: "Login successful",
      token,
      student: {
        id: student._id,
        name: student.name,
        email: student.email,
      },
    });
  } catch (error) {
    return res.status(500).json({ message: "Server error", error: error.message });
  }
});

module.exports = router;
```

Grievance Routes

```
const express = require("express");
const Grievance = require("../models/Grievance");
const authMiddleware = require("../middleware/authMiddleware");

const router = express.Router();

router.use("/grievances", authMiddleware);

router.post("/grievances", async (req, res) => {
  try {
    const { title, description, category, status } = req.body;

    if (!title || !description || !category) {
      return res
        .status(400)
        .json({ message: "Title, description and category are required" });
    }
```

```
    const grievance = await Grievance.create({
      title,
      description,
      category,
      status: status || "Pending",
      student: req.user.id,
    });

    return res.status(201).json({ message: "Grievance submitted", grievance });
  } catch (error) {
    return res.status(500).json({ message: "Server error", error: error.message });
  }
});

router.get("/grievances", async (req, res) => {
  try {
    const grievances = await Grievance.find({ student: req.user.id }).sort({
      createdAt: -1,
    });
    return res.status(200).json(grievances);
  } catch (error) {
    return res.status(500).json({ message: "Server error", error: error.message });
  }
});

router.get("/grievances/search", async (req, res) => {
  try {
    const { title = "" } = req.query;

    const grievances = await Grievance.find({
      student: req.user.id,
      title: { $regex: title, $options: "i" },
    }).sort({ createdAt: -1 });

    return res.status(200).json(grievances);
  } catch (error) {
    return res.status(500).json({ message: "Server error", error: error.message });
  }
});

router.get("/grievances/:id", async (req, res) => {
  try {
    const grievance = await Grievance.findOne({
      _id: req.params.id,
      student: req.user.id,
    });

    if (!grievance) {
      return res.status(404).json({ message: "Grievance not found" });
    }

    return res.status(200).json(grievance);
  } catch (error) {
    return res.status(500).json({ message: "Server error", error: error.message });
  }
});
```

```
router.put("/grievances/:id", async (req, res) => {
  try {
    const { title, description, category, status } = req.body;

    const grievance = await Grievance.findOneAndUpdate(
      { _id: req.params.id, student: req.user.id },
      {
        ...(title !== undefined && { title }),
        ...(description !== undefined && { description }),
        ...(category !== undefined && { category }),
        ...(status !== undefined && { status }),
      },
      { new: true, runValidators: true }
    );

    if (!grievance) {
      return res.status(404).json({ message: "Grievance not found" });
    }

    return res.status(200).json({ message: "Grievance updated", grievance });
  } catch (error) {
    return res.status(500).json({ message: "Server error", error: error.message });
  }
});

router.delete("/grievances/:id", async (req, res) => {
  try {
    const grievance = await Grievance.findOneAndDelete({
      _id: req.params.id,
      student: req.user.id,
    });

    if (!grievance) {
      return res.status(404).json({ message: "Grievance not found" });
    }

    return res.status(200).json({ message: "Grievance deleted" });
  } catch (error) {
    return res.status(500).json({ message: "Server error", error: error.message });
  }
});

module.exports = router;
```

.env Example

```
PORT=5000
MONGO_URI=mongodb://127.0.0.1:27017/student_grievance_db
JWT_SECRET=your_super_secret_key
```

.git (Version Control)

Git repository initialized and pushed to GitHub: `https://github.com/UpwanSingh/AIFSD_MSE_2`

Frontend Code

App.jsx

```
import { Navigate, Route, Routes } from "react-router-dom";
import RegisterPage from "./pages/RegisterPage";
import LoginPage from "./pages/LoginPage";
import DashboardPage from "./pages/DashboardPage";
import ProtectedRoute from "./components/ProtectedRoute";

const App = () => {
  return (
    <Routes>
      <Route path="/register" element={<RegisterPage />} />
      <Route path="/login" element={<LoginPage />} />
      <Route
        path="/dashboard"
        element={
          <ProtectedRoute>
            <DashboardPage />
          </ProtectedRoute>
        }
      />
      <Route path="*" element={<Navigate to="/login" replace />} />
    </Routes>
  );
};

export default App;
```

main.jsx

```
import React from "react";
import ReactDOM from "react-dom/client";
import { BrowserRouter } from "react-router-dom";
import App from "./App";
import "./styles.css";

ReactDOM.createRoot(document.getElementById("root")).render(
  <React.StrictMode>
    <BrowserRouter>
      <App />
    </BrowserRouter>
  </React.StrictMode>
);
```

RegisterPage.jsx

```
import { useState } from "react";
import { Link, useNavigate } from "react-router-dom";
import { api } from "../api";

const RegisterPage = () => {
  const navigate = useNavigate();
```

```
  const [formData, setFormData] = useState({
    name: "",
    email: "",
    password: "",
  });
  const [message, setMessage] = useState("");
  const [error, setError] = useState("");

  const handleChange = (e) => {
    setFormData((prev) => ({ ...prev, [e.target.name]: e.target.value }));
  };

  const handleSubmit = async (e) => {
    e.preventDefault();
    setError("");
    setMessage("");

    try {
      const response = await api.post("/register", formData);
      setMessage(response.data.message || "Registration successful");
      setTimeout(() => navigate("/login"), 800);
    } catch (err) {
      setError(err.response?.data?.message || "Registration failed");
    }
  };

  return (
    <div className="page-wrapper">
      <div className="card">
        <h2>Student Registration</h2>
        <form onSubmit={handleSubmit} className="form-grid">
          <input type="text" name="name" placeholder="Name" value={formData.name}
onChange={handleChange} required />
          <input type="email" name="email" placeholder="Email" value={formData.email}
onChange={handleChange} required />
          <input type="password" name="password" placeholder="Password" value=
{formData.password} onChange={handleChange} required />
          <button type="submit">Register</button>
        </form>
        {message && <p className="success-text">{message}</p>}
        {error && <p className="error-text">{error}</p>}
        <p>Already registered? <Link to="/login">Login</Link></p>
      </div>
    </div>
  );
};

export default RegisterPage;
```

LoginPage.jsx

```
import { useState } from "react";
import { Link, useNavigate } from "react-router-dom";
import { api, setAuthToken } from "../api";

const LoginPage = () => {
```

```
  const navigate = useNavigate();
  const [formData, setFormData] = useState({ email: "", password: "" });
  const [error, setError] = useState("");

  const handleChange = (e) => {
    setFormData((prev) => ({ ...prev, [e.target.name]: e.target.value }));
  };

  const handleSubmit = async (e) => {
    e.preventDefault();
    setError("");
    try {
      const response = await api.post("/login", formData);
      const { token, student } = response.data;
      localStorage.setItem("token", token);
      localStorage.setItem("student", JSON.stringify(student));
      setAuthToken(token);
      navigate("/dashboard");
    } catch (err) {
      setError(err.response?.data?.message || "Invalid login");
    }
  };

  return (
    <div className="page-wrapper">
      <div className="card">
        <h2>Student Login</h2>
        <form onSubmit={handleSubmit} className="form-grid">
          <input type="email" name="email" placeholder="Email" value={formData.email}
onChange={handleChange} required />
          <input type="password" name="password" placeholder="Password" value=
{formData.password} onChange={handleChange} required />
          <button type="submit">Login</button>
        </form>
        {error && <p className="error-text">{error}</p>}
        <p>New user? <Link to="/register">Register</Link></p>
      </div>
    </div>
  );
};

export default LoginPage;
```

DashboardPage.jsx

```
import { useEffect, useMemo, useState } from "react";
import { useNavigate } from "react-router-dom";
import { api, setAuthToken } from "../api";

const initialForm = {
  title: "",
  description: "",
  category: "Academic",
  status: "Pending",
};
```

```
const DashboardPage = () => {
  const navigate = useNavigate();
  const [grievances, setGrievances] = useState([]);
  const [formData, setFormData] = useState(initialForm);
  const [searchTitle, setSearchTitle] = useState("");
  const [editingId, setEditingId] = useState(null);
  const [error, setError] = useState("");
  const [success, setSuccess] = useState("");

  const student = useMemo(() => {
    const raw = localStorage.getItem("student");
    return raw ? JSON.parse(raw) : null;
  }, []);

  const handleUnauthorized = () => {
    localStorage.removeItem("token");
    localStorage.removeItem("student");
    setAuthToken(null);
    navigate("/login");
  };

  const fetchAllGrievances = async () => {
    try {
      const response = await api.get("/grievances");
      setGrievances(response.data);
    } catch (err) {
      if (err.response?.status === 401) {
        setError("Unauthorized access");
        handleUnauthorized();
      } else {
        setError("Failed to fetch grievances");
      }
    }
  };

  useEffect(() => {
    const token = localStorage.getItem("token");
    if (!token) {
      navigate("/login");
      return;
    }

    setAuthToken(token);
    fetchAllGrievances();
  }, []);

  const handleChange = (e) => {
    setFormData((prev) => ({ ...prev, [e.target.name]: e.target.value }));
  };

  const clearMessages = () => {
    setError("");
    setSuccess("");
  };

  const resetForm = () => {
    setFormData(initialForm);
```

```
    setEditingId(null);
  };

  const handleSubmit = async (e) => {
    e.preventDefault();
    clearMessages();

    try {
      if (editingId) {
        await api.put(`/grievances/${editingId}`, formData);
        setSuccess("Grievance updated successfully");
      } else {
        await api.post("/grievances", formData);
        setSuccess("Grievance submitted successfully");
      }

      resetForm();
      fetchAllGrievances();
    } catch (err) {
      if (err.response?.status === 401) {
        setError("Unauthorized access");
        handleUnauthorized();
      } else {
        setError(err.response?.data?.message || "Action failed");
      }
    }
  };

  const handleDelete = async (id) => {
    clearMessages();
    try {
      await api.delete(`/grievances/${id}`);
      setSuccess("Grievance deleted successfully");
      fetchAllGrievances();
    } catch (err) {
      if (err.response?.status === 401) {
        setError("Unauthorized access");
        handleUnauthorized();
      } else {
        setError("Delete failed");
      }
    }
  };

  const handleEdit = (grievance) => {
    setEditingId(grievance._id);
    setFormData({
      title: grievance.title,
      description: grievance.description,
      category: grievance.category,
      status: grievance.status,
    });
  };

  const handleSearch = async () => {
    clearMessages();
    try {
```

```
      const response = await api.get(`/grievances/search?
title=${encodeURIComponent(searchTitle)}`);
      setGrievances(response.data);
    } catch (err) {
      if (err.response?.status === 401) {
        setError("Unauthorized access");
        handleUnauthorized();
      } else {
        setError("Search failed");
      }
    }
  };

  const handleLogout = () => {
    localStorage.removeItem("token");
    localStorage.removeItem("student");
    setAuthToken(null);
    navigate("/login");
  };

  return (
    <div className="dashboard-wrapper">
      <div className="top-bar">
        <h2>Student Grievance Dashboard</h2>
        <div className="top-bar-right">
          <span>{student?.name || "Student"}</span>
          <button onClick={handleLogout} className="btn-secondary">Logout</button>
        </div>
      </div>
    </div>
  );
};

export default DashboardPage;
```

Frontend .env Example

```
VITE_API_BASE_URL=http://localhost:5000/api
```

---

4. Screenshots of Login, Register, Dashboard and All Functional Modules

Login, Register, Dashboard, CRUD and Logout

Student Registration

Name

Email

Password

Register

Already registered? Login

Student Login

Email

Password

Login

New user? Register

Student Login

Email

Password

Login

New user? Register

Student Grievance Dashboard

Auto Student

Logout

Submit Grievance

Title

Description

Academic

Pending

Submit

Search Grievances

Search by title

Search

Reset

All Grievances

No grievances found.

Student Grievance Dashboard

Auto Student | Logout

Submit Grievance

Title

Description

Academic

Pending

Submit

Search Grievances

Search by title | Search | Reset

Grievance submitted successfully

All Grievances

Mess issue 1776939371371

Food quality needs improvement.

Category: Hostel

Status: Pending

Date: 4/23/2026, 3:46:15 PM

Update | Delete

Student Grievance Dashboard

Auto Student | Logout

Submit Grievance

Title

Description

Academic

Pending

Submit

Search Grievances

Mess | Search | Reset

All Grievances

Mess issue 1776939371371

Food quality needs improvement.

Category: Hostel

Status: Pending

Date: 4/23/2026, 3:46:15 PM

Update | Delete

Student Grievance Dashboard

Auto Student | Logout

Submit Grievance

Title

Description

Academic

Pending

Submit

Search Grievances

Mess | Search | Reset

Grievance updated successfully

All Grievances

Mess issue 1776939371371

Food quality needs improvement.

Category: Hostel

Status: Resolved

Date: 4/23/2026, 3:46:15 PM

Update | Delete

Student Grievance Dashboard

Auto Student | Logout

Submit Grievance

Title

Description

Academic

Pending

Submit

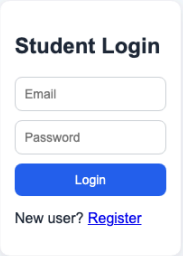
Search Grievances

Mess | Search | Reset

Grievance deleted successfully

All Grievances

No grievances found.



Postman API Testing Screenshots

UPWAN SINGH's Workspace

Search

Invite Upgrade

POST Get data

No environment

My Collection > Get data

Save Share

COLLECTIONS

My Collection

GET Get data

POST Post data

POST https://student-grievance-backend-mfqb.onrender.com/api/register Send

Docs Params Authorization Headers 8 Body Scripts Tests Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

```
1 {
2   "name": "Upwan Singh",
3   "email": "upwan_test1@gmail.com",
4   "password": "123456",
5   "role": "student"
6 }
```

Body Cookies Headers 15 Test Results 0/1 201 Created 1.28 s 562 B Save Response

Raw Preview Visualize


```
1 {"message":"Registration successful","student":{"id":"69e9f7061519a0f9d523198a","name":"Upwan Singh","email":"upwan_test1@gmail.com"}}
```


ENVIRONMENTS

SPECS

FLOWS

Connect Git Console Terminal Globals Vault Tools

UPWAN SINGH's Workspace

Search

Invite Upgrade

POST Get data

No environment

My Collection > Get data

Save Share

COLLECTIONS

My Collection

GET Get data

POST Post data

POST https://student-grievance-backend-mfqb.onrender.com/api/login Send

Docs Params Authorization Headers 8 Body Scripts Tests Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

```
1 {
2   "email": "upwan_test1@gmail.com",
3   "password": "123456"
4 }
```

Body Cookies Headers 15 Test Results 1/1 200 OK 1.28 s 758 B Save Response

Raw Preview Visualize

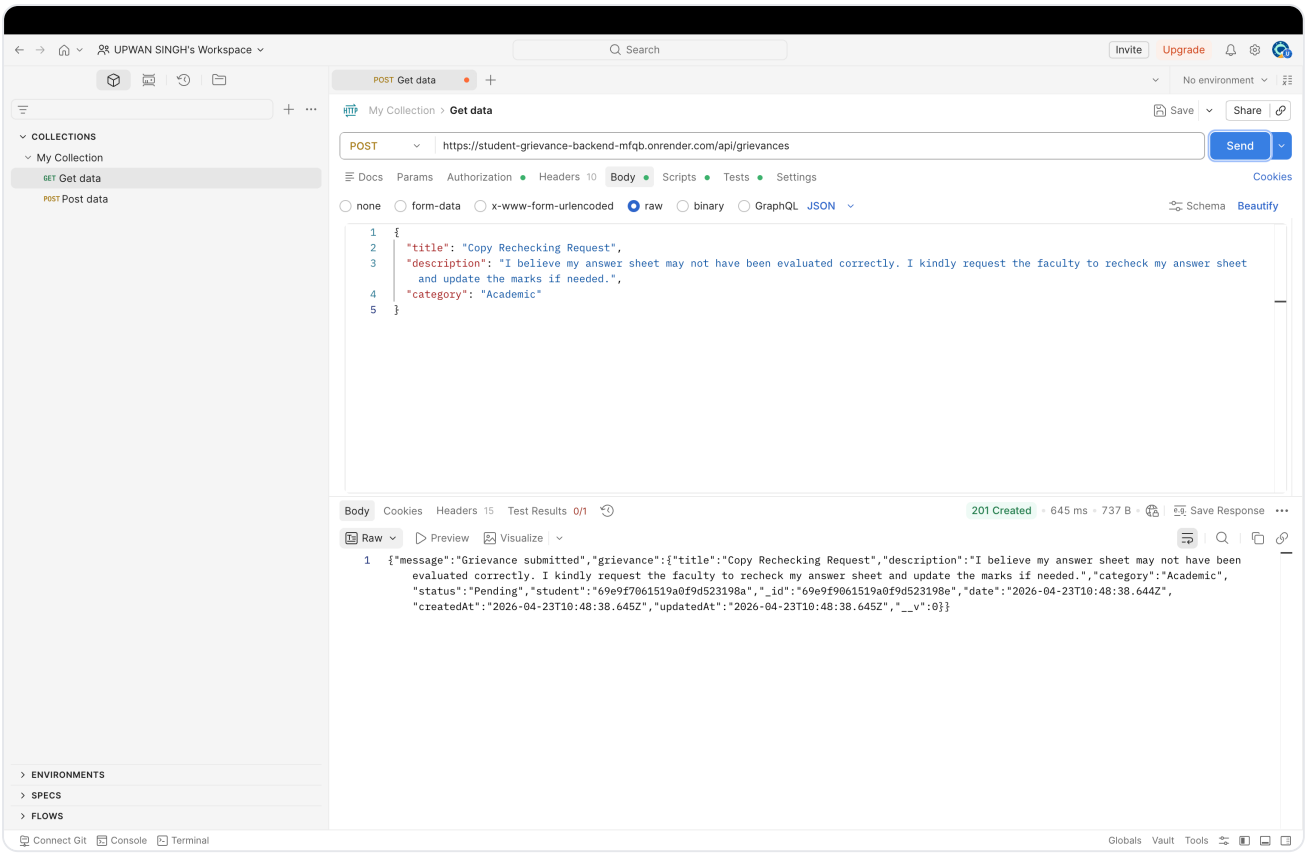

```
1 {"message":"Login successful","token":"eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.
    eyJpZCI6IjY5ZTlmNzA2MTUxOWEwZjlkNTIzMTk4YSIsIm5hbWUiOiJVcHdhbiBTaW5naCIsImVtYWlsIjoidXB3YW5fdGVzdDFAZ21haWwuY29tIiwiaWF0IjoxNzc2OTQwODI3LCJ
    leHAiOjE3NzcwMjcyMjd9.xC-tvRI-CsjJID91OtjetPjHOqSWUtFkbGBvi_tMllE","student":{"id":"69e9f7061519a0f9d523198a","name":"Upwan Singh",
    "email":"upwan_test1@gmail.com"}}
```


ENVIRONMENTS

SPECS

FLOWS

Connect Git Console Terminal Globals Vault Tools



Live API testing evidence (Render)

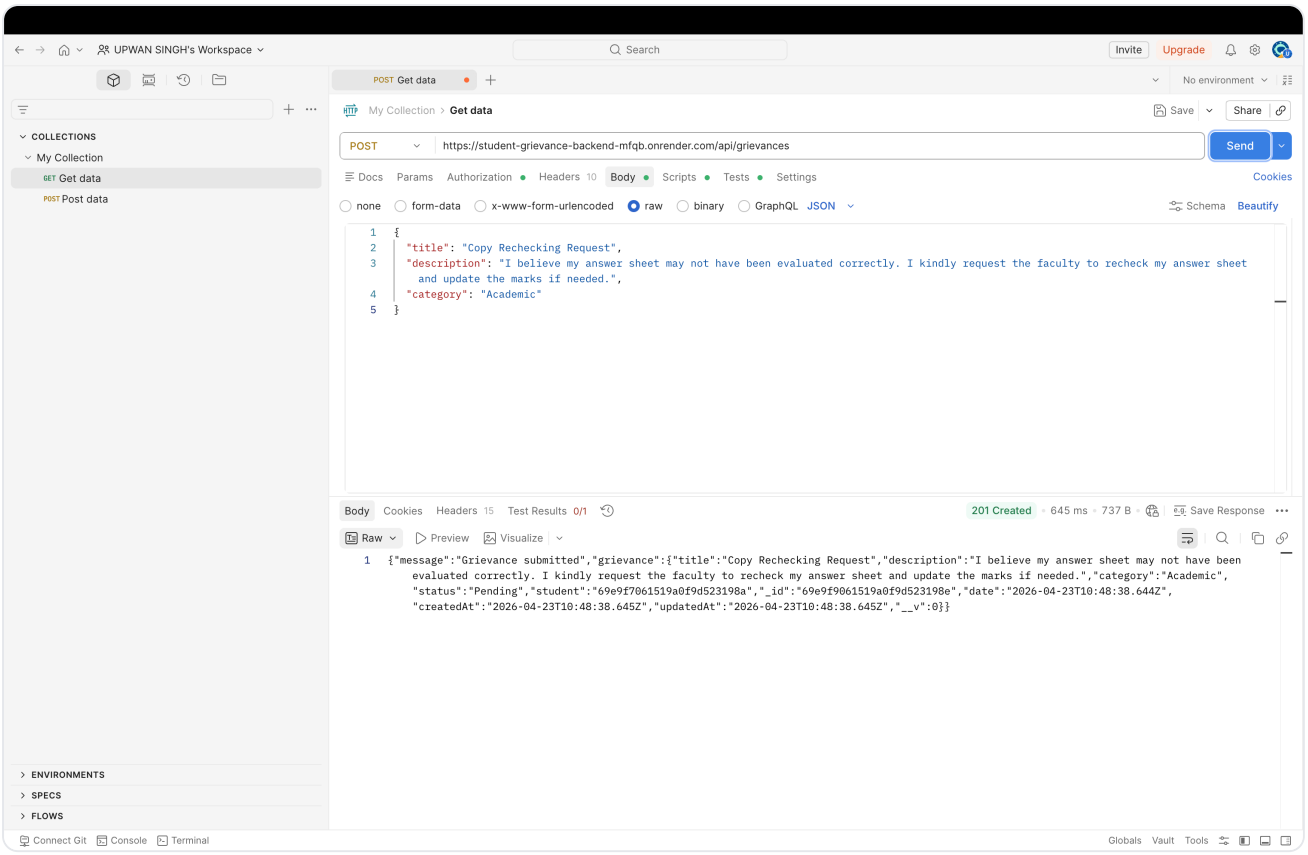
See full tested request/response output in:

- evidence/LIVE_RENDER_API_TEST_RESULTS.md

---

5. Screenshot of VS Code Project Structure

VS Code Project Structure Screenshot



Project structure shown includes backend, frontend, evidence, render.yaml, and README.md.